LAB # 5

Yusuf Bozdağ 33840 Lab Section : C

1. Introduction

In this experiment, the fundamental characteristics and practical applications of semiconductor diodes were investigated. Diodes are essential nonlinear circuit elements that allow current to flow primarily in one direction, making them critical components in rectification and voltage regulation circuits. Understanding their current–voltage (I–V) behavior is essential for designing and analyzing electronic systems.

The first part of the experiment focused on observing and comparing the I–V characteristics of different diode types, including a standard pn-junction diode, a Schottky diode, and a Zener diode. These components exhibit distinct electrical behaviors such as different forward voltage drops and breakdown mechanisms, which directly influence their applications in circuits.

In the later stages, these diode properties were utilized to construct and analyze rectifier circuits. A half-wave rectifier was implemented to demonstrate basic AC-to-DC conversion, followed by a full-wave bridge rectifier to improve output efficiency. Finally, a regulated rectifier circuit incorporating a capacitor filter and a Zener diode was built to achieve a more stable DC output with reduced ripple.

This experiment aimed to bridge the gap between theoretical diode models and real-world circuit behavior by combining characterization measurements with practical circuit implementations.

2. Pre-Lab Work

For the pre-lab work, the provided study material on diodes was reviewed in order to understand the basic concepts required for this experiment. In particular, the structure of a diode, the concepts of forward bias and reverse bias, and the general I–V characteristics of diodes were studied. These topics were examined to better understand how diodes operate and how they are used in electronic circuits.

A diode is a semiconductor device that allows current to flow mainly in one direction. It has two terminals called the anode and the cathode. When the anode is at a higher potential than the cathode, the diode becomes forward biased and allows current to flow through the device. In contrast, when the cathode is at a higher potential than the anode, the diode is reverse biased and ideally blocks the current. Therefore, diodes are commonly used to control the direction of current flow in electronic circuits.

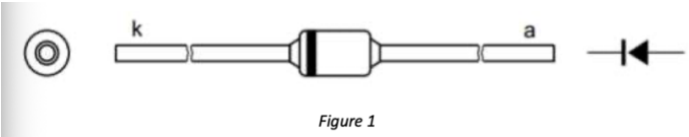


Figure 1

Analysis of Figure 2

To determine the transfer characteristic of the circuit in Figure 2, the ideal diode model is considered. In this model, a forward-biased diode behaves as a short circuit, while a reverse-biased diode behaves as an open circuit. Therefore, the behavior of the circuit depends on whether diodes D_1 and D_2 are conducting or not for different values of the input voltage.

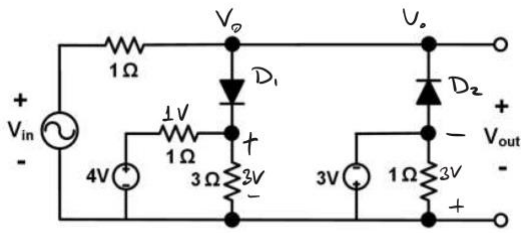


Figure 2

The left branch of the circuit establishes a reference voltage through the 4 V source and the resistor network. This creates an effective reference level of approximately +3 V for the lower terminal of diode D_1. As a result, diode D_1 remains reverse biased when the output voltage is less than 3 V and becomes forward biased when the output voltage exceeds 3 V.

Similarly, the right branch creates a reference level of −3 V due to the orientation of the 3 V source. Diode D_2 becomes forward biased when the output voltage drops below −3 V and remains reverse biased when the output voltage is greater than −3 V.

Based on these conditions, the circuit operates in three regions. When the input voltage is between −3 V and 3 V, both diodes are reverse biased and behave as open circuits, so the output follows the input voltage. When the input voltage becomes smaller than −3 V, diode D_2 turns on and clamps the output voltage to approximately −3 V. When the input voltage becomes greater than 3 V, diode D_1 turns on and limits the output voltage to around +3 V.

Figure 2

D_1 is open circuit when $V_o < 3V$, short circuit when $V_o > 3V$

D_2 is short circuit when $V_o < -3V$, open circuit when $V_o > -3V$

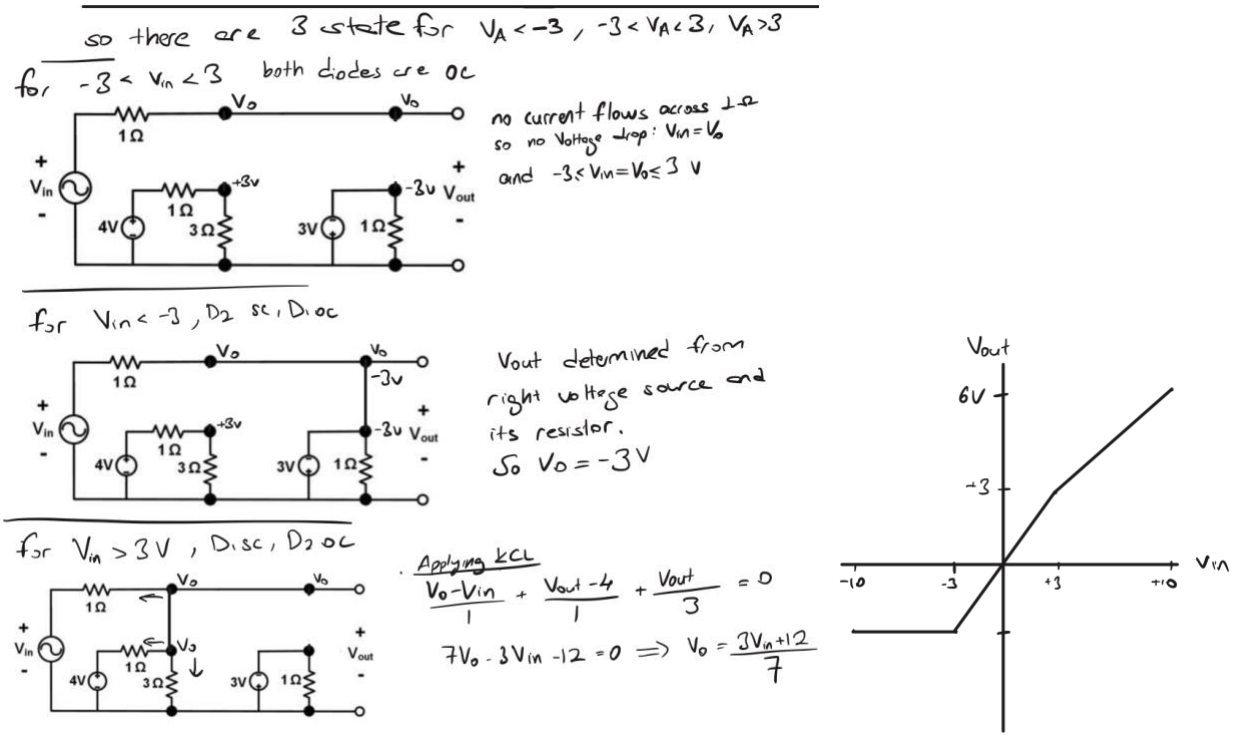


Figure 3. Hand calculations and operating states of the circuit for different input voltage regions.

For the analyzed circuit, three operating regions are obtained depending on the input voltage. When $V_{in} < -3\,V$, diode D_2 conducts and the output is clamped at $V_{out} = -3\,V$. For $-3\,V \leq V_{in} \leq 3\,V$, both diodes are off and the output follows the input ($V_{out} = V_{in}$). When $V_{in} > 3\,V$, diode D_1 conducts and the output becomes $V_{out} = (3V_{in} + 12)/7$.

This type of circuit operates as a diode limiter (or voltage clipping circuit). Its purpose is to restrict the output voltage within a specified range by using diodes and reference voltages. Such circuits are commonly used in electronic systems for signal protection, voltage limiting, and waveform shaping to prevent signals from exceeding safe voltage levels.

3. Data/Results

Part 1 – Diode I–V Characterization

Building Block

In this part of the experiment, the I–V characteristics of different types of diodes were investigated using a simple series circuit consisting of a diode and a resistor. A DC voltage source was used, and measurements were taken manually by adjusting the input voltage step-by-step.

The circuit was constructed on a breadboard with the diode connected in series with a resistor (R = 1 kΩ). Instead of using an oscilloscope, measurements were performed using a multimeter. First, the voltage across the diode (Vd) was adjusted to match the specified values in the table by varying the input

voltage. Then, the multimeter was reconfigured as an ammeter and connected in series to measure the corresponding diode current (Id). This method allowed direct observation of the relationship between diode voltage and current.

Components used:
1 kΩ resistor
1N4148 switching diode
1N5822 Schottky diode
1N5229 Zener diode
DC power supply
Digital multimeter (used as voltmeter and ammeter)
Breadboard and connection cables

The measurements were performed by manually adjusting the input voltage to obtain specific diode voltage (Vd) values listed in the table. For each Vd value, the corresponding current (Id) was measured by connecting the multimeter in series as an ammeter.
This step-by-step measurement approach enabled the construction of the I–V characteristic curves for each diode type.

Part 1.A – 1N4148 Fast Switching Diode

In this part, the I–V characteristic of the 1N4148 fast switching diode was measured experimentally. The circuit was built on the breadboard using a 1 kΩ series resistor and the diode under test. To obtain the data points, the DC source was adjusted gradually until the multimeter, used in voltmeter mode, showed the required diode voltage listed in the table. After fixing that voltage value, the multimeter was reconnected in series as an ammeter to measure the corresponding diode current. This procedure was repeated for each voltage point so that the current–voltage relationship of the diode could be recorded step by step.

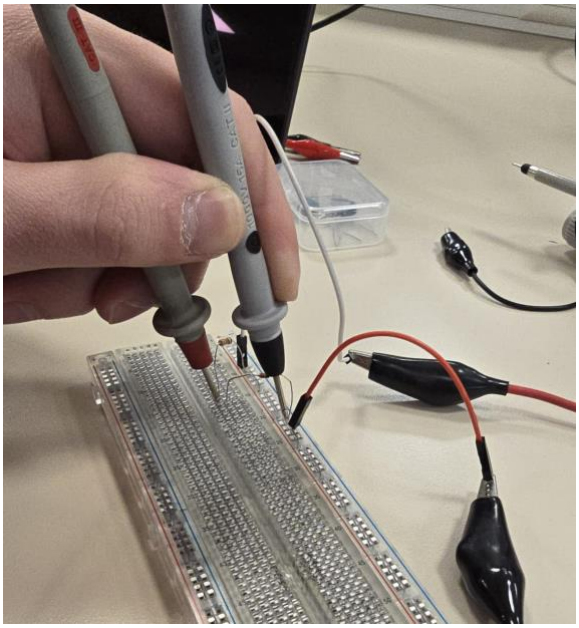


Figure 4. Breadboard implementation of the 1N4148 diode I–V measurement setup

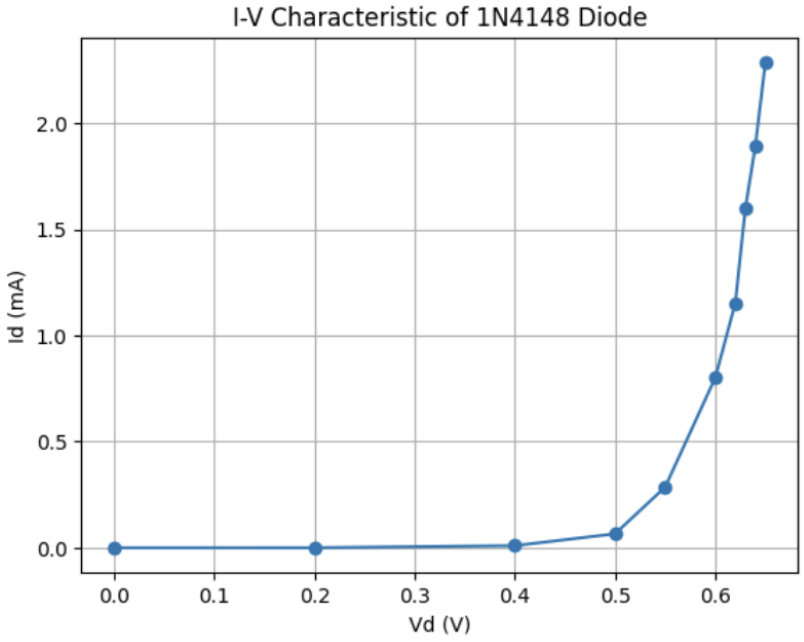


Figure 4.1 Measured voltage values of the 1N4148 diode at different operating points using the DC power supply and multimeter.

Vd (V)	0	0.2	0.4	0.5	0.55	0.6	0.62	0.63	0.64	0.65
Id (µA)	0	0.197	9.98	65.67	286.07	801.4	1153.16	1598.51	1894.11	2287.76

Table 1 Measured Data – 1N4148 Diode

These results show that the current remains very small at low forward-bias voltages and then increases rapidly as the diode voltage approaches about 0.6 V. This is the expected exponential behavior of a silicon pn-junction diode.



Graph 1

Therefore, the measured data is consistent with the expected behavior of a 1N4148 diode. The diode stays nearly non-conducting at low voltages, while significant conduction starts around 0.6 V and above. Therefore, the experimental results agree well with the theoretical I–V characteristic and with the typical forward-bias behavior given in the datasheet for a silicon switching diode.

Part 1.B – 1N5822 Schottky Diode

In this part, the I–V characteristic of the 1N5822 Schottky diode was measured experimentally. The same procedure used in Part 1.a was followed, where a DC source and a series resistor were used to control the voltage across the diode and measure the corresponding current. However, due to the relatively thick legs of the Schottky diode, it could not be directly inserted into the breadboard. Therefore, one leg of the diode was connected by wrapping it around the resistor lead, and the other leg was connected to the DC supply cable using alligator clips. This allowed a stable electrical connection while maintaining the required circuit configuration.

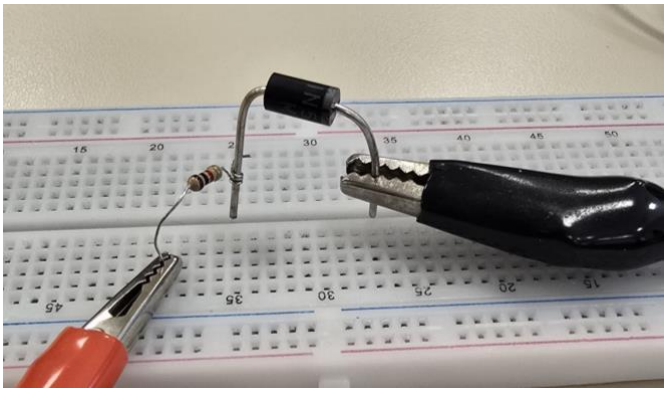


Figure 5. Experimental setup of the 1N5822 Schottky diode I–V measurement



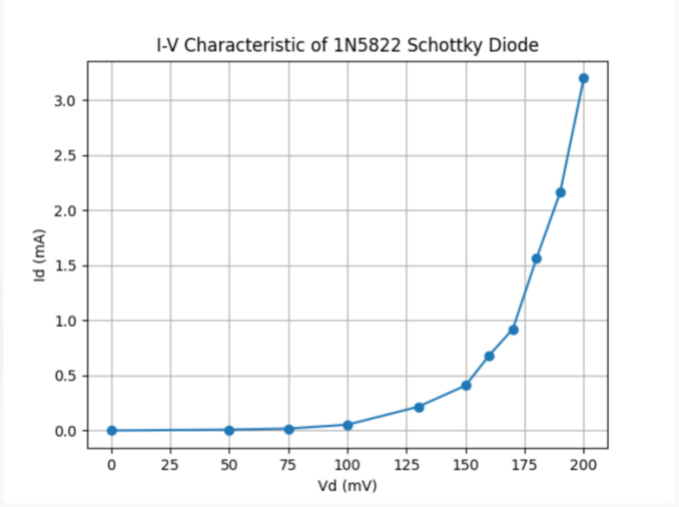
Figure 5.1 Measured voltage values of the 1N4148 diode at low forward-bias levels, showing the gradual increase in diode voltage during the initial conduction region.

The input voltage was swept in the range of 0 mV to 200 mV, and the current through the diode was recorded for each voltage value. The measured data is given in Table 2.

Vd (mV)	**0**	**50**	**75**	**100**	**130**	**150**	**160**	**170**	**180**	**190**	**200**
Id (mA)	0	0.0082	0.018	0.053	0.216	0.409	0.683	0.922	1.564	2.162	3.2

Table 2 Measured Data – 1N5822 Schottky Diode

From the measured results, it can be observed that the current starts increasing at much lower voltages compared to the silicon diode in Part 1.a. Even at around 100–150 mV, the diode begins to conduct noticeably, which is a characteristic feature of Schottky diodes. The current then increases rapidly with a small increase in voltage, indicating the exponential I–V relationship.



Graph 2

The I–V characteristic of the 1N5822 diode shows that the forward-bias threshold voltage is significantly lower than that of a standard silicon diode, typically around 0.2–0.3 V. This behavior is consistent with the theoretical expectations and datasheet specifications of Schottky diodes, which are known for their low forward voltage drop and fast switching properties.
Therefore, the experimental results confirm that the 1N5822 Schottky diode exhibits earlier conduction and higher current at lower voltages compared to the 1N4148 diode, making it suitable for applications requiring low power loss and high-speed operation.

Part 1.C – 1N5229 Zener Diode

In this part, the I–V characteristic of the 1N5229 Zener diode was measured experimentally. The circuit was built on the breadboard using a 1 kΩ series resistor and the Zener diode under test. As in the previous parts, the DC source was gradually adjusted until the desired diode voltage shown in the table was obtained, and the corresponding current was then measured using the multimeter. Since the Zener diode was characterized in both reverse-bias and forward-bias regions, measurements were taken for negative and positive diode voltages. This made it possible to observe both the normal forward conduction behavior and the reverse breakdown behavior of the Zener diode.

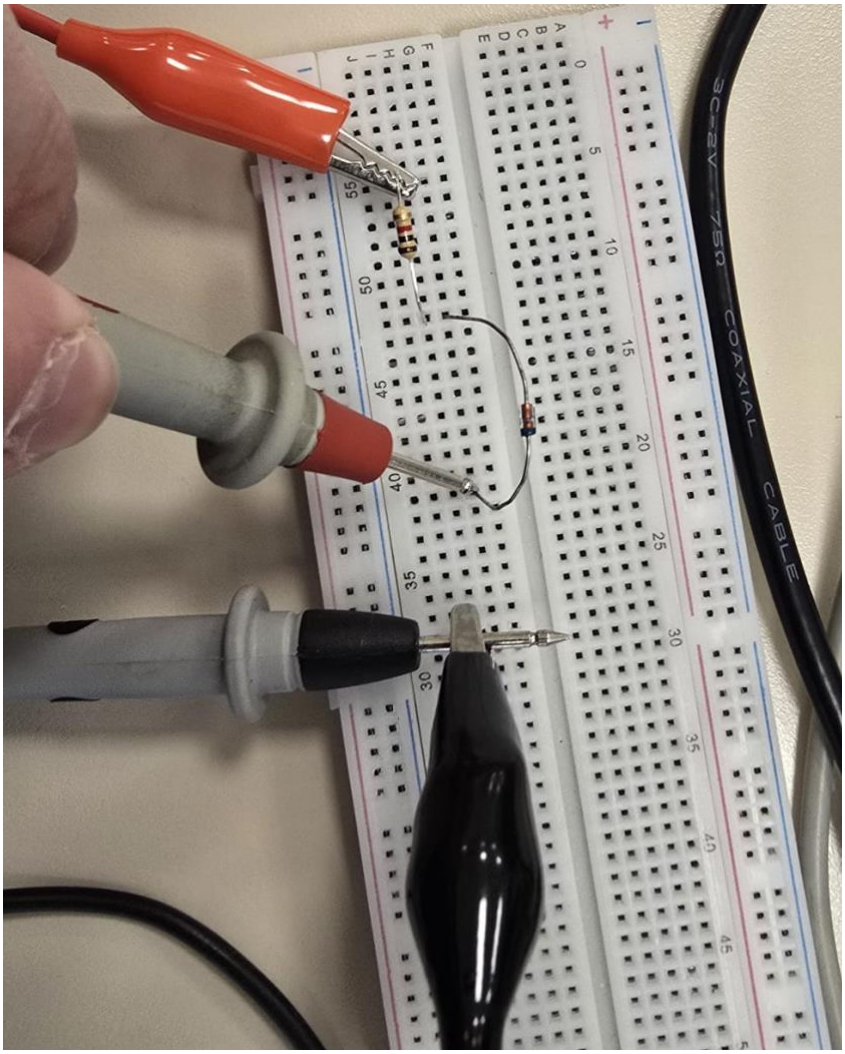


Figure 6. Breadboard implementation of the 1N5229 Zener diode I–V measurement setup

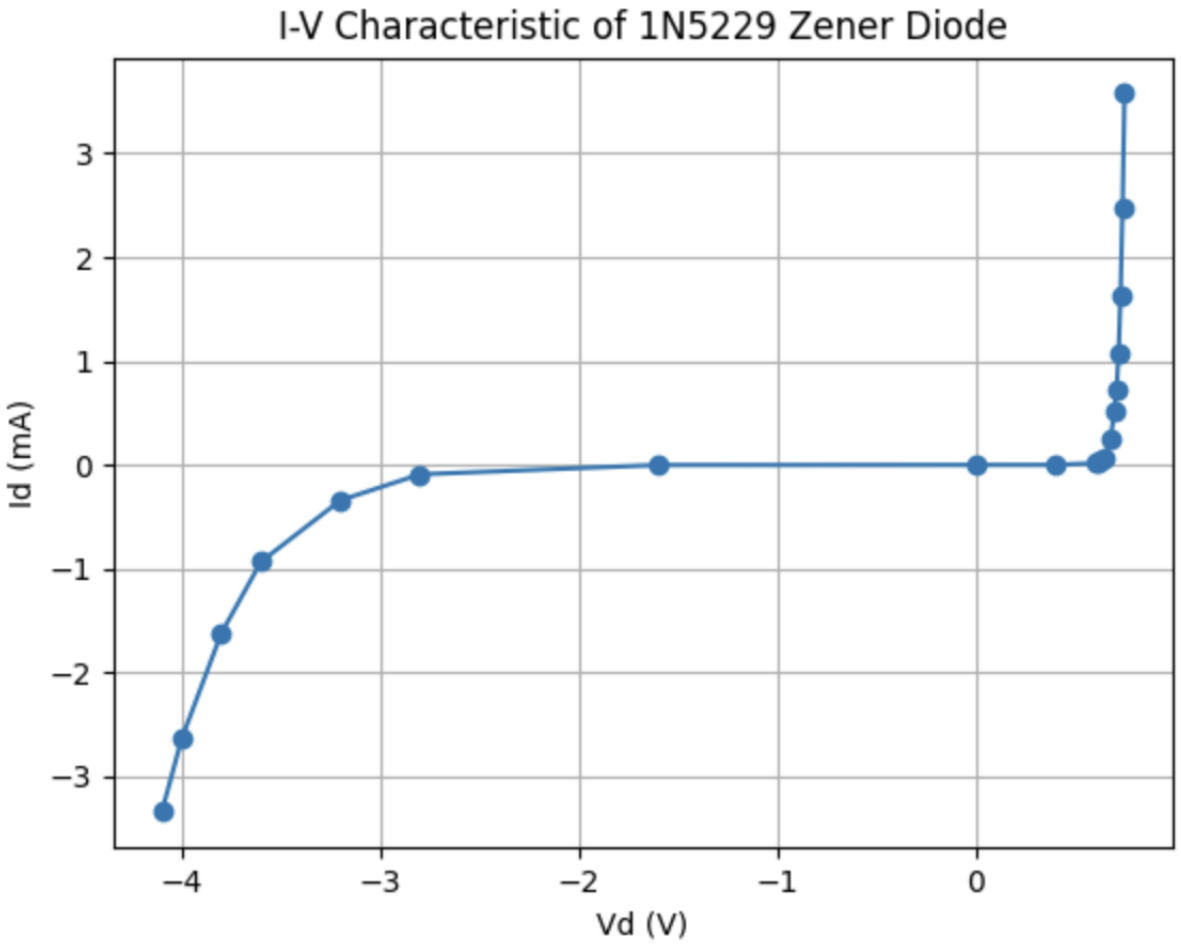


Figure 6.1 Measured voltage values of the 1N5229 Zener diode in both forward and reverse bias regions

Vd (V)	-4.1	-4.0	-3.8	-3.6	-3.2	-2.8	-1.6	0	0.4	0.6
Id (mA)	-3.34	-2.638	-1.62	-0.932	-0.341	-0.091	-0.00135	0	0.000326	0.0167
Vd (V)	0.62	0.63	0.65	0.68	0.70	0.71	0.72	0.73	0.74	0.75
Id (mA)	0.0278	0.0376	0.0599	0.261	0.527	0.731	1.071	1.639	2.483	3.578

Table 3 Measured Data – 1N5229 Zener Diode

The measured results show that in the forward-bias region, the Zener diode behaves similarly to a regular silicon diode. The current remains very small at lower positive voltages and then starts to increase rapidly after approximately 0.68–0.7 V. In the reverse-bias region, the current stays very small at first, but then increases significantly as the diode voltage approaches about -4 V. This indicates the onset of Zener breakdown. Therefore, the experimental data matches the expected behavior of a Zener diode: normal diode conduction in forward bias and breakdown-based voltage regulation in reverse bias.



Graph 3

The measured forward voltage and Zener voltage are consistent with the expected datasheet behavior of the 1N5229 diode.

The main objective of this experiment was to investigate the I–V characteristics of different types of diodes (1N4148 silicon diode, 1N5822 Schottky diode, and 1N5229 Zener diode) and compare their behaviors with theoretical expectations. The results showed that the silicon diode begins to conduct around 0.6–0.7 V, the Schottky diode conducts at lower forward voltages with faster response and lower losses, and the Zener diode behaves like a normal diode in forward bias while exhibiting breakdown at a specific reverse voltage (Zener voltage) for voltage regulation. The experiment successfully demonstrated the distinct electrical characteristics of each diode type and their suitability for different applications.

Part 2 – Half-Wave Rectifier

In this part, a half-wave rectifier circuit was built and analyzed in order to observe how an AC signal can be converted into a DC-like output. A half-wave rectifier allows only one half (positive or negative) of the AC waveform to pass through while blocking the other half, resulting in a pulsating DC signal at the output.

The circuit consists of a single diode (D1) and a load resistor (RL = 470 Ω). The diode acts as a switching element: it conducts during the positive half-cycle of the input signal and blocks current during the negative half-cycle. As a result, the output voltage across the load resistor follows only the positive portion of the input waveform.

Components used:
1 × 1N4148 diode
1 × 470 Ω resistor
Breadboard
Function generator (AC source)
Oscilloscope
Connecting wires and probes

The function generator was configured to produce a 100 Hz sine wave with an amplitude of 5 Vpp. The circuit was implemented on a breadboard as shown in Figure 9. The input signal was applied to the diode, and the output voltage was measured across the load resistor using the oscilloscope.

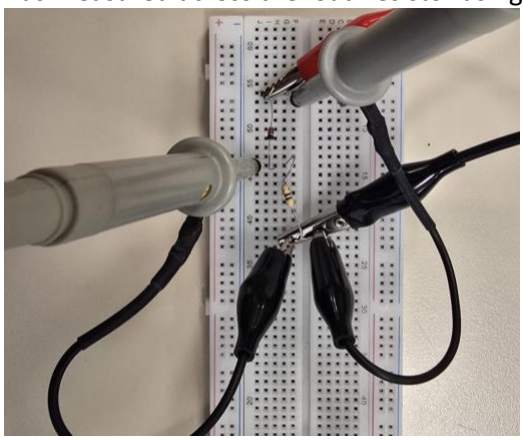


Figure 7. Breadboard implementation of the half-wave rectifier circuit with diode and load resistor.

The input and output waveforms were observed simultaneously on the oscilloscope, as shown in Figure 10. The input signal is a standard sinusoidal waveform, while the output signal consists only of the positive half-cycles. During the positive half-cycle, the diode is forward-biased and conducts, allowing current to pass through the load resistor. During the negative half-cycle, the diode is reverse-biased and blocks current, resulting in zero output voltage.

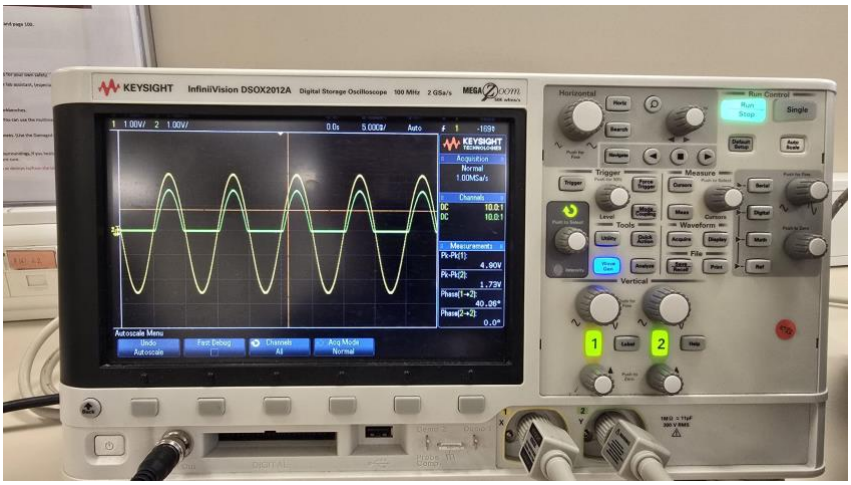


Figure 8. Oscilloscope display of input (sinusoidal) and output (half-wave rectified) signals.

It was observed that the output waveform does not reach the exact peak of the input signal due to the forward voltage drop of the diode (approximately 0.6–0.7 V for a silicon diode). Therefore, the peak of the output signal is slightly lower than the input peak.

Finally the half-wave rectifier successfully converts the AC input into a pulsating DC output by removing the negative half-cycles of the signal. This experiment demonstrates the basic principle of rectification and highlights the role of the diode as a unidirectional current device.

Part 3 – Full-Wave Bridge Rectifier

In this part, a full-wave bridge rectifier circuit was built in order to convert both the positive and negative halves of the AC input signal into a unidirectional output. Unlike the half-wave rectifier in Part 2, this circuit uses four diodes arranged in a bridge configuration, allowing current to pass through the load resistor during both half-cycles of the input waveform. As a result, the output becomes a pulsating DC signal with a higher average value and less interruption than the half-wave case.

Components used:
4 × 1N4148 diodes
1 × 470 Ω load resistor
Breadboard
Function generator
Oscilloscope
Connecting wires, probes, and alligator clips

The function generator was adjusted to produce a 100 Hz sine wave with 5 Vpp amplitude, as required in the lab manual. The bridge rectifier circuit was then implemented on the breadboard using four diodes and one load resistor, as shown in Figure 11.

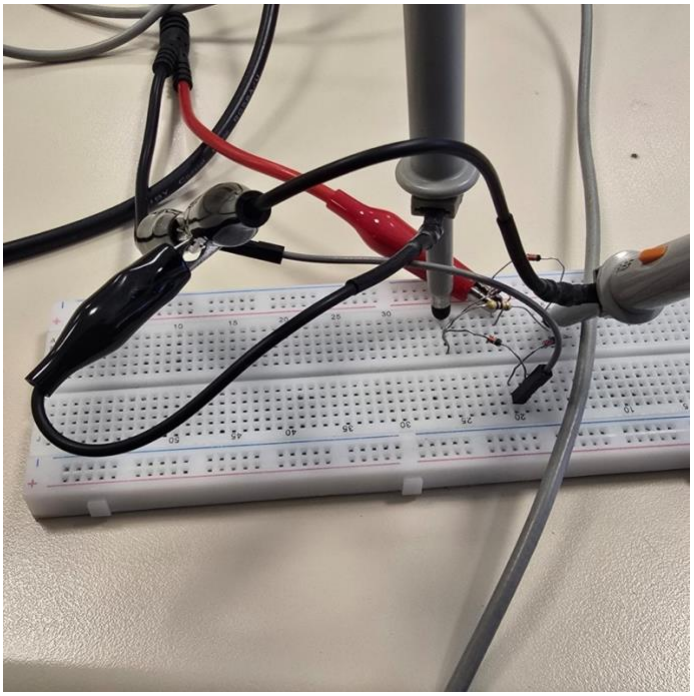


Figure 11. Breadboard implementation of the full-wave bridge rectifier circuit using four diodes and a 470 Ω load resistor.

In this part, the output voltage across the load resistor could not be measured directly with a single oscilloscope channel because neither side of the resistor was at the oscilloscope ground reference. Therefore, the voltage at each terminal of the load resistor was measured separately. One oscilloscope probe was connected to one side of the resistor, and the second probe was connected to the other side of the resistor, while both probe grounds were connected to the common ground of the circuit. Then, the oscilloscope math function was used to calculate the voltage difference across the resistor as Channel 2 − Channel 1, which gave the actual output voltage. This method allowed the output waveform to be observed safely and correctly without shorting any part of the circuit.

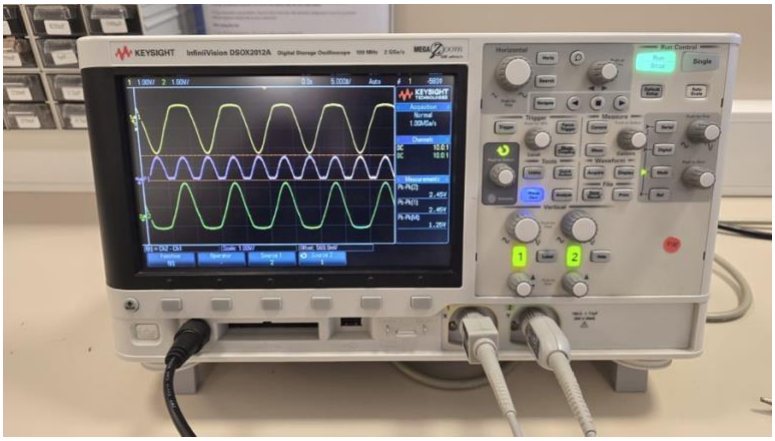


Figure 12. Oscilloscope display showing the two measured node voltages and the output voltage obtained using the math function.

The observed result confirms the expected behavior of a full-wave bridge rectifier. During the positive half-cycle of the input signal, one pair of diodes conducts, and during the negative half-cycle, the other pair conducts. In both cases, current flows through the load resistor in the same direction. Because of this, the negative half of the input waveform is effectively flipped upward, and the output consists of positive pulses for both halves of the input cycle. Compared to the half-wave rectifier, the output waveform is smoother and has twice the ripple frequency, since rectification occurs in both half-cycles. It was also observed that the output amplitude is lower than the input amplitude. This is expected because, in a bridge rectifier, current passes through two diodes in series during each half-cycle, which causes approximately two forward voltage drops. Therefore, the measured output peak is smaller than the input peak. Even with this reduction, the full-wave bridge rectifier provides a more efficient AC-to-DC conversion than the half-wave rectifier because it uses the entire input waveform rather than only one half.

This part demonstrated that a bridge rectifier produces a more continuous DC-like output than a half-wave rectifier. The oscilloscope measurements and the math-based voltage subtraction method clearly showed how both halves of the AC signal were converted into a unidirectional waveform across the load resistor.

Part 4 – Regulated Full-Wave Bridge Rectifier

In this part, the full-wave bridge rectifier circuit from Part 3 was further improved by adding a capacitor filter and a Zener diode regulator in order to obtain a more stable and nearly constant DC output. While the full-wave rectifier produces a pulsating DC signal, it still contains noticeable ripple. Therefore, a capacitor is used to smooth the output, and a Zener diode with a series resistor is used to regulate the voltage.

The circuit consists of a full-wave bridge rectifier (4 diodes), a smoothing capacitor (C = 470 µF), a series resistor (R = 100 Ω), a Zener diode, and a load resistor (R_load = 470 Ω). The AC input signal (50 Hz, 5 Vpp) is first rectified by the diode bridge, then filtered by the capacitor, and finally regulated by the Zener diode.

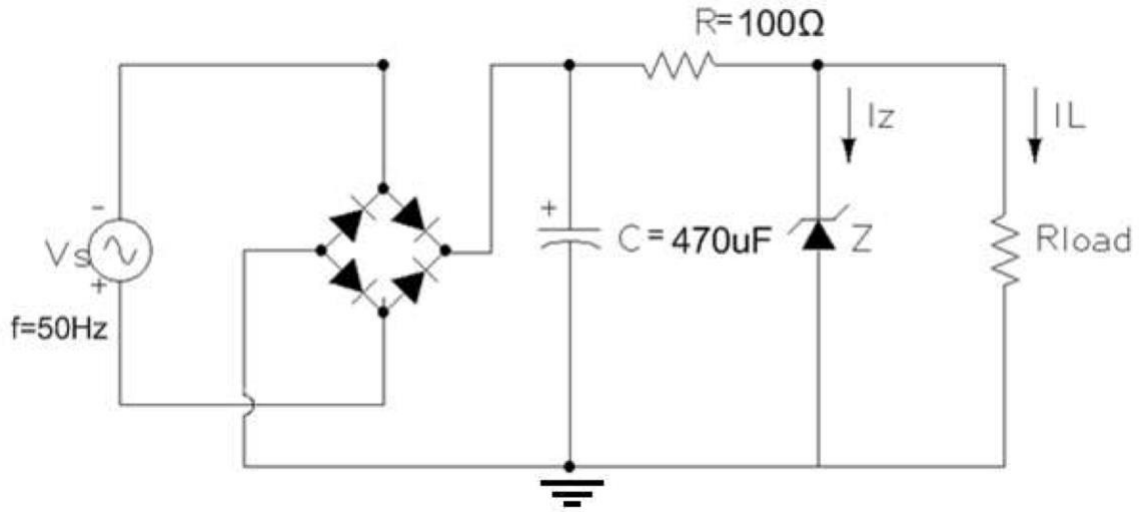


Figure 13. Circuit diagram of the regulated full-wave bridge rectifier including capacitor filter and Zener diode regulation.

The capacitor plays a key role in reducing ripple. After rectification, the capacitor charges up to the peak voltage and discharges slowly through the load when the input voltage decreases. This results in a smoother output voltage compared to the unfiltered case. The Zener diode operates in reverse breakdown region and maintains a nearly constant voltage across the load, thus providing voltage regulation even when the input or load conditions change.

Components Used:
Function Generator / AC Source
4 × Diodes (1N4148) – bridge rectifier
Capacitor (C = 470 µF) – filter capacitor
Series Resistor (R = 100 Ω)
Zener Diode (e.g., 5.1 V Zener diode)
Load Resistor
Oscilloscope (for waveform observation)
Breadboard and connection wires

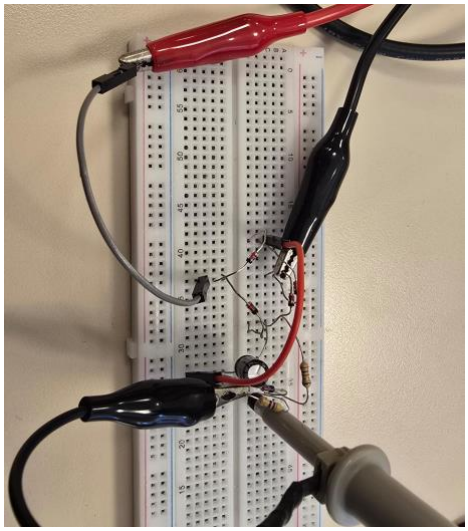


Figure 14. Breadboard implementation of the regulated full-wave bridge rectifier circuit.

For measurements, the oscilloscope was used to observe the DC output level. The waveform appears nearly flat compared to previous parts, indicating that the ripple has been significantly reduced. When the input amplitude was set to 5 Vpp, the measured DC output voltage was approximately 1.9 V under no-load conditions. When the input amplitude was reduced to 4 Vpp, the output voltage dropped to around 0.9 V, showing the effect of input variation on the regulated output.

To evaluate the performance of the circuit, load regulation and source regulation were calculated. The load regulation was calculated using the following formula: Load Regulation (%) = 100 × (V_NL − V_FL) / V_NL where V_NL is the no-load output voltage and V_FL is the full-load output voltage. From the measurements:



Figure 15. Output at $V_{in} = 5\ V_{pp}$: no-load ≈ 1.9 V, with load (470 Ω) ≈ 880 mV.

VNL = 1.9 V
VFL = 0.90 V
Substituting these values: Load Regulation (%) = 100 × (1.9 − 0.90) / 1.9 = 52.63%

This relatively high value indicates that the circuit is sensitive to load changes, mainly due to the limited current capability and non-ideal components.

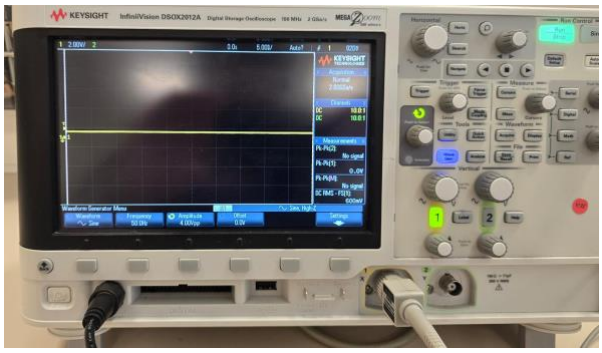


Figure 16. Output at $V_{in} = 4\ V_{pp}$, no-load condition: ≈ 600 mV.

The source regulation was calculated using the following formula: Source Regulation (%) = 100 × (Vpp5 − Vpp4) / Vpp5 where Vpp5 is the output voltage for Vin = 5 Vpp and Vpp4 is the output voltage for Vin = 4 Vpp. From the measurements:
Vpp5 = 900 mV
Vpp4 = 600 mV
Substituting these values: Source Regulation (%) = 100 × (900 − 600) / 900 = 33.3%

This shows that the output voltage also varies with changes in the input amplitude, meaning the regulation is not ideal but still improved compared to an unregulated rectifier.

This circuit illustrates how a basic AC signal can be progressively transformed into a more stable DC output by combining rectification, filtering, and regulation stages. After the bridge rectifier converts both halves of the input waveform into a unidirectional signal, the capacitor smooths the pulsating voltage by storing energy at peak levels and releasing it when the input drops. This significantly reduces fluctuations and creates a more continuous output.

The regulation stage, formed by the Zener diode and the series resistor, aims to keep the output voltage within a limited range. When the voltage exceeds the Zener breakdown level, the diode conducts and diverts excess current, preventing further increase in output voltage. In this way, the circuit attempts to maintain a controlled voltage across the load despite variations in input amplitude or load conditions. However, the measurements show that the regulation is not strong, as the output voltage changes noticeably between no-load and load conditions and also with different input amplitudes. This behavior can be explained by practical limitations of the circuit. The voltage available after the bridge rectifier is already reduced by diode drops, leaving limited headroom for proper regulation. In addition, the capacitor cannot completely eliminate ripple under load, and the Zener diode requires sufficient current to operate effectively. When the load increases, a larger portion of the current is drawn by the load instead of the Zener diode, weakening its ability to regulate the voltage.

The experiment demonstrates that while filtering and regulation improve the quality of the DC output, their effectiveness strongly depends on proper design conditions such as sufficient input voltage, appropriate resistor values, and adequate current through the Zener diode.

Bonus Lab Assignment

In this part, a biased diode clipping circuit was designed, simulated, and experimentally verified. The circuit schematic is shown in Figure 17, where an AC source (SINE(0, 2.5, 100)) is connected through a coupling capacitor (C1 = 100 nF) to a diode and a DC bias source (V1 = 3 V). A high-value load resistor (R1 = 1 MΩ) is connected at the output.

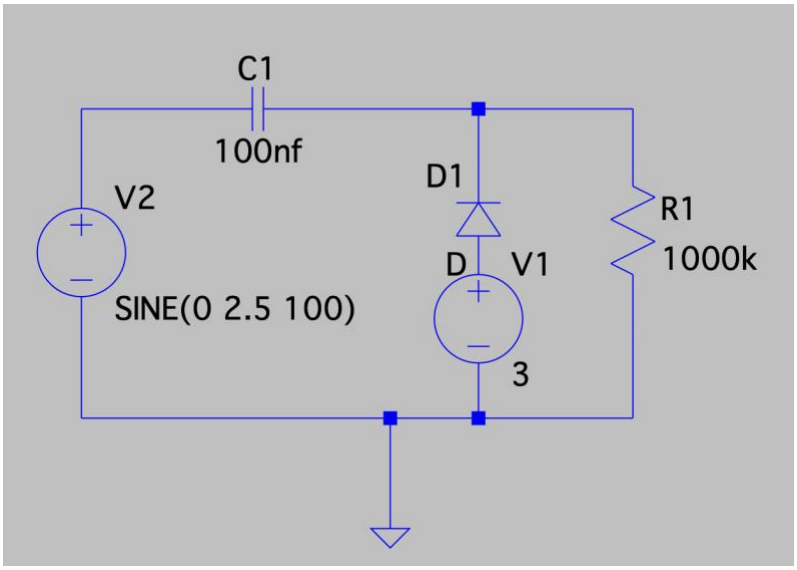


Figure 17.

Components Used

Function Generator (AC source: SINE, 2.5 V amplitude, 100 Hz)
Capacitor (C1 = 100 nF)
Diode (D1 = 1N4148)
DC Voltage Source (V1 = 3 V bias)
Resistor (R1 = 1 MΩ load resistor)
Oscilloscope

Breadboard and connection wires

In this configuration, the capacitor acts as a coupling element, blocking any DC component of the input and allowing only the AC signal to pass to the diode stage. The diode together with the 3 V DC source forms a biased clipper, which limits the output voltage when it exceeds a certain threshold. Specifically, the diode becomes forward biased only when the input signal exceeds approximately the bias voltage plus the diode forward voltage (~3 V + 0.7 V). As a result, the positive peaks of the signal are clipped, while the rest of the waveform remains unchanged.

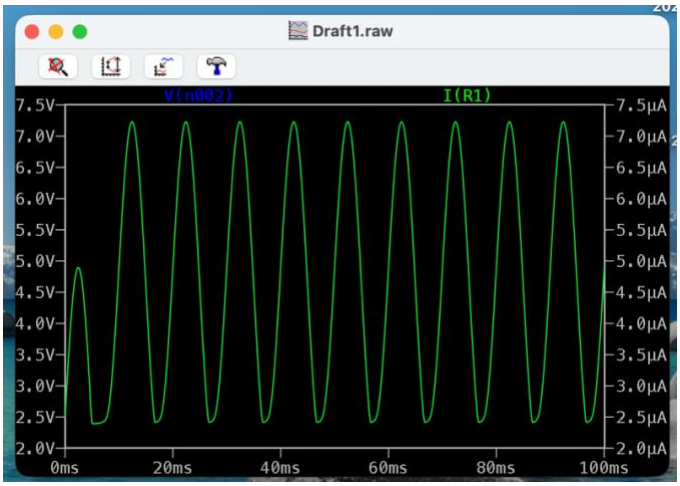
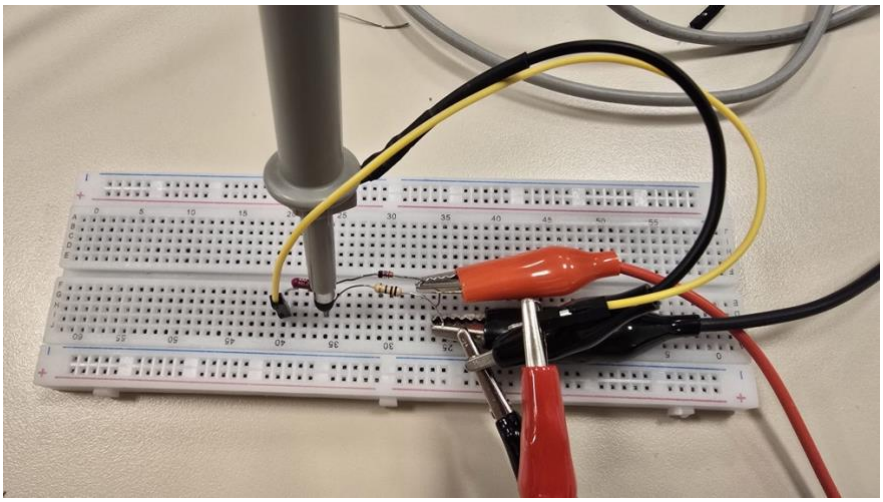


Figure 18.

The behavior of the circuit was first analyzed using LTSpice, as shown in Figure 18. The simulation output clearly demonstrates that the waveform follows the input sinusoidal shape until it reaches the clipping level, after which the peaks are flattened. This confirms that the diode conducts only beyond the defined threshold, effectively limiting the output voltage.


Figure 19.

Following the simulation, the circuit was physically implemented on a breadboard, as illustrated in Figure 19. The oscilloscope measurement of the output signal is presented in Figure 20. The observed waveform closely matches the LTSpice simulation results: the signal retains its sinusoidal form for lower amplitudes, while the upper peaks are clipped at a nearly constant level.
Small discrepancies between simulation and experimental results can be attributed to non-ideal diode behavior, such as variations in forward voltage drop, as well as parasitic elements and measurement limitations in the physical setup.

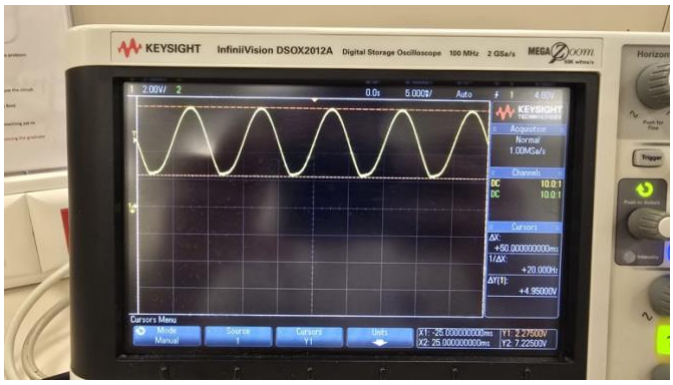


Figure 20.

This experiment demonstrates how a biased diode clipper can be used for signal conditioning and voltage limiting, and shows strong agreement between theoretical expectations, simulation, and practical implementation.

4. Discussion/Analysis

In this experiment, the behavior of different diode types and their applications in rectification, filtering, and signal shaping circuits were investigated through theoretical analysis, simulation, and experimental measurements. Overall, the results obtained from all parts are consistent with the expected theoretical behavior, although some deviations were observed due to non-ideal characteristics of real components.

In **Part 1**, the I–V characteristics of the 1N4148, 1N5822, and 1N5229 diodes were analyzed. The silicon diode (1N4148) showed a turn-on voltage around 0.6–0.7 V, while the Schottky diode (1N5822) began conducting at much lower voltages (~0.2–0.3 V), which matches theoretical expectations. The Zener diode (1N5229) behaved like a normal diode in forward bias and exhibited breakdown near its rated Zener voltage in reverse bias. The measured curves closely followed the expected exponential behavior, confirming that the experimental setup accurately captured diode characteristics. Minor differences in current values can be attributed to measurement resolution and temperature effects.

In **Part 2**, the half-wave rectifier successfully removed the negative half-cycles of the input signal, producing a pulsating DC output. The measured output waveform matched the theoretical expectation; however, the peak output voltage was lower than the input due to the forward voltage drop of the diode. This confirms that real diodes introduce voltage losses that are not present in ideal models.

In **Part 3**, the full-wave bridge rectifier improved the output by utilizing both halves of the AC signal. Compared to the half-wave rectifier, the output had a higher average value and reduced gaps between pulses, which was clearly observed in the oscilloscope measurements. However, the output amplitude

was reduced due to the presence of two diode drops in each conduction path. This result aligns with theoretical analysis and demonstrates the trade-off between efficiency and voltage loss in bridge rectifiers.

In **Part 4**, the addition of a capacitor and Zener diode significantly improved the output quality. The capacitor reduced the ripple by storing energy at peak voltages and releasing it during voltage drops, resulting in a smoother DC signal. The Zener diode provided partial voltage regulation by limiting the output voltage. However, the calculated load regulation (≈52.63%) and source regulation (≈33.3%) indicate that the regulation performance is not ideal. These deviations can be explained by several factors: insufficient input voltage headroom after diode drops, limited current through the Zener diode, and capacitor discharge under load conditions. This shows that proper regulation requires careful design of component values and operating conditions.

In the **Bonus Assignment**, the biased diode clipper circuit demonstrated waveform shaping behavior. Both LTSpice simulation and experimental results showed that the output waveform was clipped at a specific threshold determined by the bias voltage and diode characteristics. The agreement between simulation and measurement confirms the validity of the theoretical model. Small differences between the results are mainly due to the non-ideal forward voltage of the diode and parasitic effects in the breadboard setup.

Overall, the experimental results closely match theoretical predictions across all parts, confirming the fundamental principles of diode operation, rectification, filtering, and voltage regulation. The observed differences highlight the importance of considering non-ideal component behavior, voltage drops, and practical limitations when designing real electronic circuits

5. Post-Lab

In this experiment, the measured diode I–V characteristics and circuit behaviors were compared with theoretical predictions, and overall a strong agreement was observed. The silicon diode (1N4148) exhibited a forward conduction region around 0.6–0.7 V, which matches the expected behavior of a pn-junction diode. Similarly, the Schottky diode (1N5822) began conducting at lower voltages, consistent with its lower barrier potential. The Zener diode showed a clear breakdown region in reverse bias near its rated Zener voltage, confirming its voltage regulation capability.

Despite this general agreement, some discrepancies between theoretical and experimental results were observed. The measured currents were not exactly identical to ideal exponential curves, especially at low voltages. These differences can be attributed to practical limitations such as measurement accuracy, internal resistance of the instruments, temperature variations, and non-ideal diode behavior. In addition, parasitic resistances and contact imperfections in the breadboard setup may have slightly affected the results.

In the rectifier circuits, the output waveforms closely followed theoretical expectations. The half-wave rectifier successfully removed half of the input signal, while the full-wave bridge rectifier produced a more continuous output by utilizing both half-cycles. However, the output voltage levels were lower than predicted by ideal analysis due to diode forward voltage drops. In the regulated rectifier, although the capacitor significantly reduced ripple and the Zener diode provided partial stabilization, the output

voltage still varied with load and input changes. This deviation highlights the importance of sufficient current through the Zener diode and proper selection of circuit parameters for effective regulation.

The biased diode clipper circuit also showed good agreement between theory, simulation, and experimental results. The clipping level observed in both LTSpice and oscilloscope measurements matched the expected threshold defined by the bias voltage and diode forward voltage. Minor differences were again due to non-ideal component behavior and real-world limitations.

These findings are directly relevant to practical electronic circuit design. In real applications, engineers must account for non-ideal effects such as voltage drops, power losses, and component tolerances. The experiment demonstrates that while theoretical models provide a strong foundation, accurate circuit design requires consideration of real-world behavior. Overall, this study reinforces the importance of combining theory, simulation, and experimental validation when designing reliable electronic systems.

6. Conclusion

This experiment demonstrated how fundamental diode properties translate into practical circuit behavior in real electronic systems. By examining different diode types and implementing rectifier, filtering, regulation, and clipping circuits, it was confirmed that diodes effectively control current direction, shape signals, and stabilize voltages as predicted by theoretical models.

The experimental results were generally consistent with theoretical expectations. For example, the observed forward voltage levels, rectification behavior, and clipping thresholds closely matched the predicted values. However, deviations such as reduced output voltages, ripple under load, and imperfect regulation showed the impact of non-ideal factors like diode voltage drops, insufficient current for Zener operation, and component limitations.

Based on these results, the underlying theoretical assumptions and hypotheses about diode behavior and circuit operation can be considered valid, although only under practical constraints. The experiment clearly shows that while theory provides accurate predictions, real-world performance depends heavily on operating conditions and component selection.

To improve the accuracy and performance of the circuits, further experiments could be conducted by using different resistor values, higher input voltages, or more suitable Zener diode operating conditions to achieve stronger regulation. Additionally, using components with lower losses or higher precision measurement tools could help reduce discrepancies between theory and experiment.

This study highlights the importance of combining theoretical analysis with experimental validation and emphasizes that successful circuit design requires careful consideration of both ideal models and real-world limitations.